4/7/24

Economic Evaluation

Purpose of Economic evaluation:

* preparation of highway plans at the national regional & local level.
* To rank schemes within highway sector plan competing for scarce resources in order of priority
* To compare mutually exclusive schemes and select the most attractive one.
* To determine whether the scheme under consideration is worth investment at all.

To Evaluate:

* alternative strategies stage (or) full construction.
* " specification such as flexible (or) rigid pavement.
* " policies such as increased outlay of maintenance (or) rehabilitation.
* " design standard and
* " policy options on axle loads.

5/7/24 :

Aspects of Project appraisal :

* Engineering aspects (Construction)
* Managerial aspects
* Financial aspects
* Economic aspects (Cost & benefit) (*)

i) Engineering aspects :

Deal primarily with the technical construction process and operating of the project after it is completed as well as with the estimates of capital and operating costs.

ii) Managerial aspects :

Deal with the multitude of management & staffing problems involved in constructing & operating the project.

iii) Financial aspects :

Deal with the cost and revenue of enterprise responsible for project

iv) Economic aspects :

Deal with the economic costs and benefits from the point of view of the country as a whole.

Total transportation Cost

Construction Cost

Maintenance Cost

Road user Cost

Construction Cost:

* Survey, investigation and design cost
* land acquistion cost
* Construction Cost
* physical Contingencies
* Supervision, quality control & administration charges.

Maintenance Cost:

* ordinary repairs
* periodic repairs
* operation expenses
* Supervision and operational charges.

Road user Cost:

i) Vehicle operating Cost

* Fuel and lubricants Costs
* Spare part Cost
* Tyre Cost
* Maintenance labour Cost
* Fixed Cost
* Crew Cost
* Commodity Cost.
* Depreciation Cost

ii) Time Cost :

* Value of occupation time
* Value of Goods of transit
* Value of time of Commercial Vehicle.

iii) Accident Cost :

* Cost of Human Fatal accident
* Loss due to Injury
* Cost of hospitalization
* Damage to property Vehicle.

Benefits for Highway Improvements :

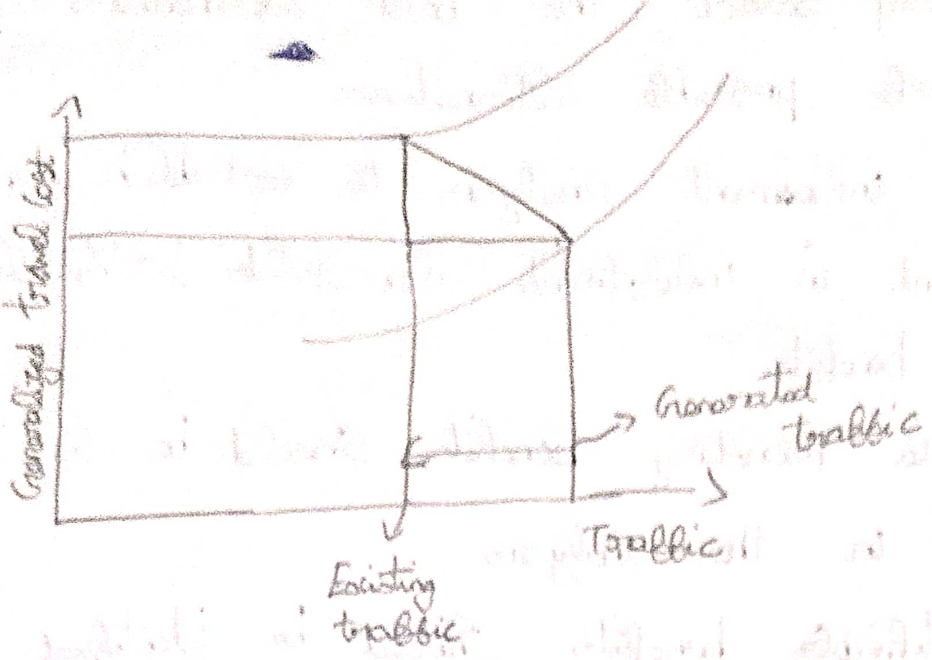
Road user benefit :

* Vehicle operating Cost Saving
* Value of travel time Savings
* Value of Savings in accident Cost
* Savings in maintenance Cost.

Quantifiable benefit :

* To existing traffic by way of reducing operating Costs, Savings in travel time and reduction in accident Costs.
* Benefits to generated traffic.
* Benefits to ~~deliver~~ diverted traffic.
* Benefits to traffic on other roads.

6/7/24 :



Non-quantifiable benefits :

* Comfort and Convenience
* Improvement in general amenities, medical services, social, educational, recreational.
* less accidents
* Improved mobility
* Good accessibility to places
* Savings enjoyed by road users transferred to other sectors.

Principles of Economic Evaluation :

* Analysis has to be done from a national viewpoint due to multiplicity of interests. Consequences to any person (or) anybody should be accounted for.
* Study of future.

* Evaluate & Select the best alternatives from a number of possible alternatives.
* Carry out incremental analysis to establish whether the increment in investment also yields justifiable incremental benefits.
* Quantifiable monetary benefits should be considered and included in the analysis.
* Non-quantifiable benefits should be identified & presented to the decision maker eg: Environmental effects
* limit analysis period to one of the reliable forecasts [Highways - 15 to 20 yrs, Expressways - upto 5 yrs]
* Omit some components of some cost & benefit which remain equal in magnitude for all alternatives.
* Devaluing costs & benefits to present value becomes necessary.

Economic & Financial Analysis:

Economic	Financial
* Concerns with the consequences to all segments of society.	* Ways & means of financing a project & financial profitability of project.
* May be rejected based on financial analysis.	* Financial Viability (Based on toll collection).

Methods of Economic Evaluation:

1) Rate of return methods

a) Benefit Cost (B/C) ratio method.

b) First year Rate of return method.

2) Discounting Cash flow (DCF) methods:

a) Net present Value (NPV) method.

b) Internet Rate of Return (IRR) method.

1) Rate of return Methods:

a) Benefit Cost (B/C) ratio method

* Widely used for evaluation of highway projects

* It is the basis of AASHTO Road user analysis.

* Ratio of net annual benefits to net annual costs is determined.

* Benefits are determined for single reference year (or) median year of analysis period.

* Costs are the equivalent annual charge representing equal amortization and interest payment spread over the economic life of the project.

$$B/C = \frac{\text{Benefits in the ref year}}{\text{Annual Cost.}}$$

Ratio = Road user Cost for the existing road − " " " " improved road in ref year
/ Equivalent annual highway Cost for the improved road − Equivalent " " " ~~Rather~~ " Existing road

1) A Single lane road 50 Km long is to be widen to 2 lane roads at a cost of 8 lakhs per Km including all improvements. The cost of operation of vehicles on the single lane road is 1.2 rupees per vehicle Km, whereas it is 1 per vehicle Km on the improved facility. The average traffic may be assume 2500 vehicles (one value then subtract cost) & two value take average per day over a design period of 20 years. The interest rate is 10% per annum. The cost of maintenance is 5,000 per Km in the existing road & 10,000 per Km in the improved roads. Is the investment in the improved scheme is worth ?

Soln:

Capital recovery factor $P = \dfrac{r(1+r)^n}{(1+r)^n - 1}$

$n \rightarrow$ design period $\qquad r = 10$

$r \rightarrow$ rate of interest $\qquad n = 20$

Annual road cost in the year

Improved road cost $= 365 \times 50 \times 2500 \times 1$

$= 456$ lakhs

Existing road cost $= 365 \times 50 \times 2500 \times 1.2$

$= 547$ lakhs.

Capital recovery factor $= \dfrac{10\%(1+10\%)^{20}}{(1+10\%)^{20} - 1} = \dfrac{\frac{10}{100}\left(1+\frac{10}{100}\right)^{20}}{\left(1+\frac{10}{100}\right)^{20} - 1}$

$= 0.117$

Total Cost of improvement = 50 × 8 lakhs
= 400 lakhs

Present annual Value = 400 lakhs × 0.117
= 46.8 lakhs.

Maintenance Cost difference = 10000 − 5000 Km
= 5000 Km.

Annual Maintenance Cost = 5000 × 50
= 2.5 lakhs

$$B/C \text{ Ratio} = \frac{547 - 455}{(46.8) + 2.5} = 1.86$$

B/C Ratio 1.86 > 1 ∴ improved road is worth.

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2) It is proposed to widen a stretch of a single lane road of length 40 Km to 2 lanes at a total cost of 125 lakhs per Km and the rate of interest is 10% per year. The annual cost of maintenance of the existing single lane road is Rupees 21,000 per Km and 75,000 per Km. The average vehicle operation cost is in the existing road is ₹4 per vehicle Km & The widen road is estimated to be ₹3 per vehicle Km. If the present traffic is 6000 motor vehicle per day & by the end of 15 yrs 2 vehicles average designed period the traffic is estimated to be doubled. Determine whether the investment & on the improvement is economically viable buyable during the 15 yrs period.

Soln:

Average traffic during design period $= \frac{6000 + 12000}{2}$ → doubled

$= 9000$.

Existing road cost $= 365 \times 40 \times 9000 \times 4$

$= 5256$ lakhs

Improved road cost $= 365 \times 40 \times 9000 \times 3$

$= 3942$ lakhs.

Benefit of year $= 5256 - 3942$

$= 1314$ lakhs.

Capital recovery $= \frac{r(1+r)^n}{(1+r)^n - 1}$

$r = 10\% = \frac{10}{100} = 0.1$

$n = 15$

Capital recovery $= \frac{0.1(1+0.1)^{15}}{(1+0.1)^{15} - 1}$

$= \frac{0.1 \times 4.177}{4.177 - 1} = \frac{0.4177}{3.177} = 0.1314$

Total cost of improvement $= 40 \times 125$ lakhs

$= 5000$ lakhs.

Present annual value $= 5000 \times 0.1314$

$= 657$ lakhs

Maintenance Cost = ~~Rs~~ 75000 - 21000

= 54000

Annual Maintenance Cost = 54000 × 40
= 2160000

Total Annual Cost = 657 lakhs + 2160000
= 678 lakhs.

$$\text{B/C Ratio} = \frac{1314 \text{ lakhs}}{678 \text{ lakhs}} = 1.938$$

B/C Ratio = 1.938.

11/7/24 :

b) First year of Return Method :

* Followed by Dept of Environment, UK.
* Benefits accrued during the first year of the scheme's operation alone are compared with the capital costs of construction.
* Results are expressed as benefits accruing in 1st yr as percentage of costs.

$$\text{FYRR} = \frac{\text{Benefits accruing in 1st year}}{\text{Capital Costs}}.$$

* FYRR of possible scheme gives an indication of its priority when compared with other schemes.

2) Discounted Cash flow Methods :

a) Net present Value (NPV) Method :

* Costs / benefits associated with the project over an extended period of the time is calculated & discounted at a selected discount rate to give Present value.

* Benefits are treated as +ve values & costs are treated as -ve values.

* If the NPV value obtained is +ve, then acceptable
-ve not acceptable.

* NPV is algebrically expressed as

$$NPV_0 = (B_0 - C_0) + \frac{B_1 - C_1}{(i+1)^2} + \frac{B_2 - C_2}{(i+1)^2} + \dots + \frac{B_n - C_n}{(i+1)^n}$$

$NPV_0 \rightarrow$ NPV in yr 0.

B_t = Value of benefits which occur in year 't'.

C_t = Cost which occur in year 't'.

i = discount rate per annum.

n = No. of years for which the return is to be calculated.

* While comparing more than 1 project, project with the highest NPV is accepted.

b) Internet Rate of Return (IRR) Method :

* It is the discount rate which makes the discounted future benefits equal to initial outlay.

* In otherwords discount rate which makes the stream of cash flows to zero.

$$C_0 = \frac{B_1 - C_1}{(i+1)^1} + \frac{B_2 - C_2}{(i+1)^2} + \dots \frac{B_n - C_n}{(i+1)^n}$$

$$= \sum_{t=1}^{n} \frac{B_t - C_t}{(i+1)^t}$$

* Tedious procedure as only trial & error can be performed computer programme makes its simpler.

* If rate of return > interest obtainable by investing the capital in open market, scheme is acceptable.

Comparison of Various Methods of Economic Evaluation:-

a) B/C ratio :

* Assumption of rate of interest which should be somewhat related to opportunity cost of capital which is not known often can be estimated only approximately.

* Significance of B/C ratio is ambiguous, difficult to understand & interpret.

* Benefits & costs are different to determine.

b) First year Rate of Return:

* Quick to use
* Some have attractive benefits initially but then tapper off.

c) Net present Value:

* Discount rate has to be assumed initially.
* Simpler in Computation compared to IRR.

d) Internal Rate of Return:

* Avoids Selection of a discount rate initially.
* Rate derived can be compared with Market rates of interest with which the economists and financial experts are familiar.
* Thus, it is more meaningful than the others.
* Disadvantages ⇒ Calculations are tedious, Sometimes misleading in comparing projects having different lives and streams of benefits.

12/07/24:

A new bypass is to be constructed at the busy town. The length of the bypass will be 5.2 Km & length of the road through the town is 5.4 Km. The cost of the project likely to be 75 lakhs. The speed of traffic through the town is 46.6 KPH. The predicted traffic after completing a bypass is 7,600 Vehicles per day, out of which 50% will use the bypass. It is Computed that if the bypass is not Constructed the Speed through the town will further drop to 44.1 KPH due to increased traffic & the Speed to the bypass is expected to be 77 KPH and that through the town will be 50.4 KPH. The travel Cost at the three speeds

Speed (KPH)	Travel Cost per Vehicle/Km
44.1	1.14
50.4	1.02
77	0.90

It is expected that the Construction of bypass will bring down the accident rate from 1.75 per million Vehicle Km on the existing route to 0.6 per million vehicle Km on the bypass. The Cost of accident can be taken as Rs 15,000. The maintenance cost per km is Rs.10,000. Calculate the first year rate of return.

<u>Soln:</u>

		Existing road	Bypass	Existing road left open after Construction of bypass
~~Ann~~ Travel Cost				
i) Travel Cost of vehicle/Km	=	1.14	0.92	1.02
Annual Cost	=	$365 \times 5.4 \times 7600 \times 1.14$	$= 365 \times 5.2 \times 7600 \times 0.9 \times \frac{50}{100}$	$= 365 \times 5.4 \times 7600 \times 1.02 \times \frac{50}{1000}$
		$= 17076744$	$= 6491160$	$= 7639596$

2) Accident cost (J) per Million | 1.75 | 0.6 | 1.75.

Accident rate per million vehicle/km $= \frac{365 \times Vol \times length \times J}{1000000}$

Accident rate/vehicle $= \frac{365 \times 5.4 \times 7600 \times 1.75}{1000000}$ $\quad \frac{365 \times 5.2 \times 7600 \times 0.6 \times 0.5}{1000000}$ $\quad \frac{365 \times 5.4 \times 7600 \times 1.75 \times 0.5}{1000000}$

$= 26.2 \quad = 4.3 \quad = 13.1$

3) Extra Maintenance Cost $= 5.2 \times 10000$
$= 52,000.$

Net benefit:

i) Savings in travel cost $= $

$$\begin{array}{r} 6491160 \\ 7639596 \quad (+) \\ \hline 14130756 \\ \hline \end{array}$$

$$\begin{array}{r} 17076744 \\ 14130756 \quad (-) \\ \hline 2945988 \\ \hline \end{array}$$

ii) Savings in Accident rate $= 26.3 - (4.3 + 13.1)$

$= 8.9 \times 15000$

$= 133500.$

Net benefits = 2945988

$$\begin{array}{r} 2945988 \\ 133500\ (+) \\ \hline 3079488 \\ \hline \end{array}$$

Maintenance Cost = 52000

$$\begin{array}{r} 3079488 \\ 52000\ (-) \\ \hline 3027488 \\ \hline \end{array}$$

First yr rate of return $= \dfrac{3027488}{7500000} \times 100$

$= 40.3\%$

18/7/24

A Section of road is at present having poor geometrics. The Cost of upgrading this road is Rs. 100 lakhs. The road user Cost with and without the improvements & the extra Maintenance Cost with the improvements are tabulated below for 20yrs period after the execution of improvement program. Assuming a discount rate 10% per annum, is the project economically justified?.

Benefit = (Col 3 − Col 2) + (Col 5 − Col 4)

$$NPV = B_0 - C_0 + \frac{B_1 - C_1}{(1+r)^{n}}$$

Year	Road Cost With improvements	Road Cost Without improvements	Accident Cost With improvement	Accident Cost without improvement	Extra Cost of Maintenance	Benefits	Benefit Cost	NPV
0	—	—	—	—	—	—	-100	-100
1	105.5	126.5	1.1	3.1	1.0	23.0	22	20
2	110.3	132.2	1.1	3.1	1.0	24.1	23.1	19.6
3	115.8	138.9	1.2	3.5	1.0	25.4	24.4	18.3
4	121.6	145.8	1.2	3.7	1.0	26.7	25.7	17.5
5	127.6	153.0	1.3	3.8	1.0	27.9	26.9	16.7
6	134.0	161.0	1.3	4.0	1.0	29.7	28.7	16.2
7	140.7	168.9	1.4	4.2	1.0	31.0	30.0	15.4
8	147.8	177.0	1.5	4.4	1.0	32.1	31.1	14.5
9	155.1	186.2	1.6	4.7	1.0	34.2	33.2	14.1
10	162.9	195.2	1.6	4.9	1.5	35.6	34.1	13.1
11	171.0	205.0	1.7	5.1	1.5	37.4	35.9	12.6
12	179.6	214.6	1.8	5.4	1.5	38.6	37.1	11.8
13	188.9	226.1	1.9	5.7	1.5	41.3	39.8	11.5
14	198.0	237.5	2.0	5.9	1.5	43.4	41.9	11.0
15	207.9	248.6	2.1	6.2	1.5	44.5	43	10.3
16	218.3	262.1	2.2	6.6	1.5	48.2	46.7	10.2
17	229.2	274.2	2.3	6.9	1.5	49.6	48.1	9.5
18	240.7	288.5	2.4	7.2	1.5	52.6	51.1	9.2
19	252.7	303.0	2.5	7.6	1.5	55.4	53.9	8.8
20	256.3	318.0	2.7	8.0	1.5	58.0	56.5	8.4

268.2 − 100 = 168.2

Since NPV is +ve the project is economically justified.

20/7/24

VoC

3 Compo

i) Vehicle

ii) Trav

iii) Acc

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i) Vari

* Fuel

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* * * * * * * *

20/7/24

Unit-2

Vehicle operating Cost (VOC):

3 Components: (Road user Cost)

i) Vehicle operating Cost.
ii) Travel time Cost.
iii) Accident Cost.

VOC types:

i) Variable Cost:

* Fuel
* Lubricants - Engine oil - other oils - Grease
* Tyre
* Spare parts
* Maintenance labour Cost
* Depreciation.

ii) Fixed Cost:

* Interest on Capital
* Insurance
* Road tax
* other taxes
* Registration fee
* Garaging charges
* permit Charges
* other fixed Charges Such as fines, tolls, octroi
* loading & unloading charges
* Commission on booking.

Overhead charges :

* Rent & water
* Clerical & Managerial staff
* Audit fees
* Advertisement
* Electricity
* Telephone & postage
* Stationary & printing
* Travelling expenses
* Freight & hire
* legal fees
* Bank charges
* Repairs & Maintenance of buildings
* Depreciation of fixed assets
* Sundries.

22/7/24 :

Cost of operation of a passenger Bus in India (1990)

A. Variable Costs			
S.NO	Component		Cost in ₹ Per Km
1	Fuel		0.68
2	Lubricants		
	i) Engine oil	0.11	
	ii) other oil	0.01	
	iii) Grease	0.01	
	Total 2	0.13	0.13
3	Tyre		0.73
4	Spare parts		0.19
5	Maintenance labour cost		0.08
6	Depreciation		0.19
	Total 'A'		$\boxed{2.00}$
B.	Fixed costs		
7	Crew costs		0.79
8	other fixed costs		1.02
	Total 'B'		$\boxed{1.81}$
	Total 'A' & 'B'		3.81

Garaging charges:

i) parking fees: This is the most common meaning. Garaging charges could refer to the cost of parking your car in the commercial garage, like one you might find downtown or at an airport.

ii) Storage fees: In some cases, it might refer to the cost of storing a vehicle in a garage for longer period, such as in a self-storage facility.

iii) Maintenance charges for a personal garage: less commonly, it could be reference to fees associated with maintaining a personal garage, like those included in home owner association (HOA) dues

octroi:

i) Historical context: octroi was a prevalent tax system in India during British rule. It was a way for municipalities to generate revenue for maintaining the city's infrastructure & services.

ii) Function: Goods brought into the city for consumption or sale were taxed at an octroi post, while goods leaving the city might also be subject to a tax.

iii) Abolition : The octroi system was gradually phased out in India after independence due to inefficiency & corruption. Today other levies like forms of taxes like GST & property tax are used for municipal revenue.

Permit Charges :

i) Regulatory Tool :

* permit charges act as a regulatory tool used by governments to manage the transportation sector.
* They can be imposed on various aspects like:
  -> Vehicle types
  -> Routes
  -> Entry / Exit points.

ii) Revenue Generation :

* permit charges can also be a source of revenue for governments.
* This revenue can be used to fund:
  -> Infrastructure development & Maintenance
  -> Traffic management initiatives.

25/7/24 :

Freight and hire:

Freight : It refers to good transported overland. Water (or) air.

Eg : Raw materials (coal, oil) to finished products (furniture, electronics) being moved b/w business (or) consumers.

Economic Significance : Freight Movement is vital for economic activity. Efficient freight transportation keeps supply chains moving & goods affordable.

Hire : In transport economics, "hire" typically refers to the leasing or renting of transportation services.

Eg : Companies might hire trucking companies, shipping lines (or) airlines to transport their freight. Individuals might hire taxis, rental trucks (or) rideshare services.

Legal fees :

Legal fees can come into play in a few ways:

* Regulatory Compliance : Legal fees would be incurred for consultations, drafting documents (or) representing the company in court if necessary.

* Accidents & Insurance: legal representation might be required in case of accidents involving vehicles, infrastructure damage, (or) disputes with insurers.

* Public-private partnerships (PPs): Transport infrastructure projects often involve complex contracts b/w governments & private companies.

Bank Charges:

Paying for transportation-related expenses:

* understanding your bank's fees for using your debit or credit card for transportation purchases (eg: gas stations, public transport passes) can help you budget effectively.

* International travel: If you're planning to travel internationally, be aware of potential foreign transaction fees associated with using your cards abroad.

Repairs and Maintenance of buildings:

* Costs and investments: The cost of repairs & maintenance for these buildings contributes to the overall budget of transportation projects. unexpected repairs can lead to budget overruns.

* Efficiency and Service quality : Well-Maintained buildings Contribute to efficiency & quality of transportation Services.

* Indirect effects:
  - land use & development : Transportation infrastructure often developed

Depreciation of fixed assets:

* Depreciation refers to the gradual decrease in the Value of fixed assets over time due to:
  - -> physical wear & tear
  - -> Technological obsolescence
  - -> Economic factors.

Economic importance:

* Accounting
* Investment decisions

Sundries:

* Minor operating expenses : In some cases, Sundries Might be used within a transportation Company's budget to represent Various minor operating expenses that are difficult to Categorize elsewhere.
* User - Side Miscellaneous Costs : less likely but theoretically, Sundriness could refer to miscellaneous Costs Incurred by users of transportation System that aren't directly ~~[illegible]~~ related to fares controll.

Crew Costs:

Different transportation modes rely on various crew members to ensure safe & efficient operation. Eg:

-> Aviation
-> Maritime
-> Rail
-> Trucking
-> Public transport.

* Cost Components: Crew costs encompass various elements.
* Salaries & wages: The base compensation paid to crew members based on experience, qualifications & regulations.
* Benefits: Health insurance, pensions, & other employee benefits contribute to overall crew cost.

26/7/24:

Factors affecting Vehicle operating Costs:

A) Vehicle Factors:

* Age
* Make
* Horse power, Engine Capacity
* load Carried
* Condition of Vehicle
* level of Maintenance i/p
* Types of fuel used.
* Types of tyres used.

B) Roadway Factors:

* Roughness Surface
* Type of Surface
* Horizontal Curvature
* Vertical profile
* pavement width
* Type & Condition of Shoulder
* Urban & rural location
* No. of Junctions per km.

* Roughness of Surface : It increases Vehicle operating costs due to higher fuel consumption, tire, wear & Vehicle Maintenance.

* Types of Surface : Different Surfaces have Varying levels of friction and Smoothness, affecting fuel Consumption & Vehicle Wear & tear.

* Horizontal Curvature : Roads with frequent & Sharp Curves require Vehicles to slow down & accelerate more often, leading to higher fuel Consumption & brake wear.

* Vertical profile : Roads with steep grades require more Vehicles to p

v) Pavement width : Narrow roads can cause slower speeds & more frequent braking & acceleration, increasing fuel consumption & vehicle wear.

vi) Type & condition of shoulder : poor & non-existent shoulders

vii) Urban and Rural location : Urban roads often have more stop and go traffic, traffic signals, & congestion, leading to higher fuel consumption & vehicle wear compared to rural roads.

viii) No. of Junctions per km : A higher no. of junctions leads to more frequent stops & starts increasing fuel consumption & vehicle wear.

c) Traffic Factors :

* Speed of travel
* Traffic volume & composition

D) Environmental factors :

* Altitude
* Rainfall
* temperature.

29/7/24:

C) Traffic Factors:

i) Speed of travel:

Impact: The speed at which vehicles affects ~~that~~ fuel consumption & wear & tear. Generally fuel efficiency is optimal at moderate speeds & efficiency decreases at very high (or) very low speeds.

ii) Traffic Volume and Consumption:

Impact: High traffic volumes & a mix of different types of vehicles can lead to congestion, frequent stops & varied speeds all of which increase fuel consumption & vehicle wear.

D) Environmental factors:

i) Altitude:

Impact: Higher altitudes can affect performance due to lower oxygen levels, leading to increased fuel consumption & potentially higher VOC.

ii) Rainfall:

Impact: Wet roads increase rolling resistance & the likelihood of accidents, leading to higher fuel consumption & increased vehicle maintenance cost

iii) Temperature :

Extreme temperatures both hot & cold can affect vehicle performance & fuel efficiency. Cold temperatures increase fuel consumption as engines take longer to reach optimal operating temperatures, while hot temperatures can affect air conditioning usage & engine cooling systems.

Cat ①

Value of travel time savings :

* Travel time is precious.
* Travel time savings is highly ~~related~~ relevant in economic.

*